

AERO2579 Aerodynamic Design Report for Group number 27

Hugh Williams ([REDACTED]), [REDACTED] ([REDACTED]), [REDACTED] ([REDACTED]), [REDACTED]
[REDACTED] ([REDACTED]), [REDACTED] ([REDACTED])

School of Engineering, RMIT University,
GPO Box 2476, Melbourne, Victoria 3001, Australia,

Abstract

The Design Build and Fly Project tasks engineers with combining theoretical, analytical and computational tools for the experimentation of designing a chuck glider optimised for the longest airborne time in specified test conditions. Multiple designs were evaluated within the group's parameters and test conditions with the criteria of optimal turbulent flow performance and low speed stability. The final design achieved consistent results during the flight evaluation, exhibiting consistent flight stability in a balanced state across all test runs. However, the achieved average airborne time was much lower in comparison to leading competitors at 1.67s to 5.2s, possibly unfavoured by the design choice of the elevator and tail. The effect was a greater negative rate of change of altitude than approximated, thus affecting the ability to maintain a longer glide. Despite the difference in comparable flight performance, the experimental hypothesis and design goals were mostly met within reasonable difference, demonstrating stable and repeatable flight and efficiency. For future iterations, the horizontal stabilator and elevator designs would be prioritised to improve the sink rate performance.

1) Introduction

This report documents the design experiment, where the group of engineers have been tasked with creating a hand-chuck glider with the primary objectives of achieving a steady balanced state and the maximum average airborne time possible. The group will compete against other gliders made by opposing groups participating in the Principles of Aerodynamics course at RMIT. The final marks based on the performance will be determined using the equation $M = e^{\frac{t}{0.482 \times T}} + 2$ (1), where M is the mark received, t is the time the glider was airborne and T is for the greatest airborne time of the gliders competing.

Gliders are a form of aircraft which fly without any forward mechanical thrust, instead relying on lift generation to remain airborne. To achieve the set goals, the design and fabrication is of the utmost importance. Different airfoils dictate aspects of geometry such as camber as a percentage of chord, the position of the camber in tenths of the chord, and more. The size of the airfoil is also essential when designing a glider. Identical airfoil shapes with differing chord and width dimensions will also have a different surface area (represented by A), which from $L = C_L \frac{1}{2} \rho V^2 A$ (2) and $D = C_D \frac{1}{2} \rho V^2 A$ (3), where L represents the lift force, C_L represents the coefficient of lift, ρ is the air density, V is the velocity, D is the drag force and C_D is the coefficient of drag, it is clear how an increase in the surface area proportionally increases the overall lift and drag of an aircraft.

Due to the properties of aerofoils outlined above, the group concurred that the glider design with the best potential would be one that has not only the greatest C_L relative to C_D , but also the geometric dimensions that maximise this difference with the materials available. This would give us the greatest lift-to-drag ratio possible, granting the opportunity to maximise the glider's airborne time. At the conclusion of this project, the group will have completed the full manufacturing process after the selection, fabrication and testing of a final chosen glider design, achieving a finished product which is able to maintain an exemplary amount of airtime.

The aerofoil and tail designs for the glider most suitable to the flight test conditions will be selected in addition to a complete analytical estimation and computational analysis. This will be done by evaluating various wing properties that are most functional for the glider to achieve a steady and balanced flight. The entire glider assembly is then outlined and manufactured applying the chosen properties. Following the flight competition the group will evaluate the gliders performance and results shown in subsequent sections.

2) Airfoil Analysis

To find the most suitable lifting section for the glider, 5 candidate airfoils were chosen and compared. NACA 0012, NACA 0015, NACA 2412, NACA 4412 and NACA 6412. These five candidates were chosen to provide a wide range of chamber distributions and thickness-to-cord ratios for comparison. These selections enable a comprehensive analysis of lift, drag, boundary-layer behaviour, and structural performance.

2.1 Theoretical framework

The aerodynamic performance of these selected airfoils was evaluated using a traditional two-dimensional flow theory. For the cambered airfoils, the velocity distributed over the chord is mainly impacted by the geometry of the airfoil, and the flow characteristics are determined by the Reynolds number. Standard sea level atmospheric conditions were assumed, with an air density value of $\rho = 1.225 \text{ kg/m}^3$ and dynamic viscosity of $\mu = 1.81 \times 10^{-5} \text{ kg/(m/s)}$. The analysis was conducted under a wide range of Reynolds numbers. Ranging from 200,000 to 500,000 in increments of 50,000. This range was chosen to match the glider's expected flight conditions.

The Reynolds number is calculated using:

$$Re = \frac{\rho V L}{\mu}$$

For the chord length calculations, a constant launch velocity of $V = 10 \text{ m/s}$ can be used to find the corresponding chord length for a range of Reynolds numbers. As shown below.

$$L = \frac{\mu Re}{\rho V}$$

• Assume velocity to be constant at sea level
Testing Reynolds numbers at 200,000 - 500,000 in 50,000 increments

$$Re = \frac{\rho V L}{\mu}$$

$$Re = \frac{(1.225)(10)(L)}{1.81 \times 10^{-5}}$$

$$V \cdot L = \frac{1.81 \times 10^{-5} Re}{1.225}$$

Reynolds Number	Final Possible Chord Length (m)
$200 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (200,000)}{1.225}$ $= \frac{724}{245} \approx 2.955$	$= 0.29 \approx 0.3 \text{ m}$
$250 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (250,000)}{1.225}$ $= \frac{181}{42} \approx 3.694$	$= 3.69 \approx 0.37 \text{ m}$
$300 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (300,000)}{1.225}$ $= \frac{1086}{245} \approx 4.43$	$= 4.43 \approx 0.44 \text{ m}$
$350 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (350,000)}{1.225}$ $= \frac{181}{25} \approx 5.171$	$= 5.171 \approx 0.52 \text{ m}$
$400 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (400,000)}{1.225}$ $= \frac{1448}{245} \approx 5.910$	$= 5.910 \approx 0.59 \text{ m}$
$450 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (450,000)}{1.225}$ $= \frac{1620}{245} \approx 6.648$	$= 6.648 \approx 0.66 \text{ m}$
$500 \times 10^3 : VL = \frac{1.81 \times 10^{-5} (500,000)}{1.225}$ $= \frac{362}{42} \approx 7.388$	$= 7.388 \approx 0.73 \text{ m}$

Figure 1: Chord Length calculations in a range of Reynolds numbers

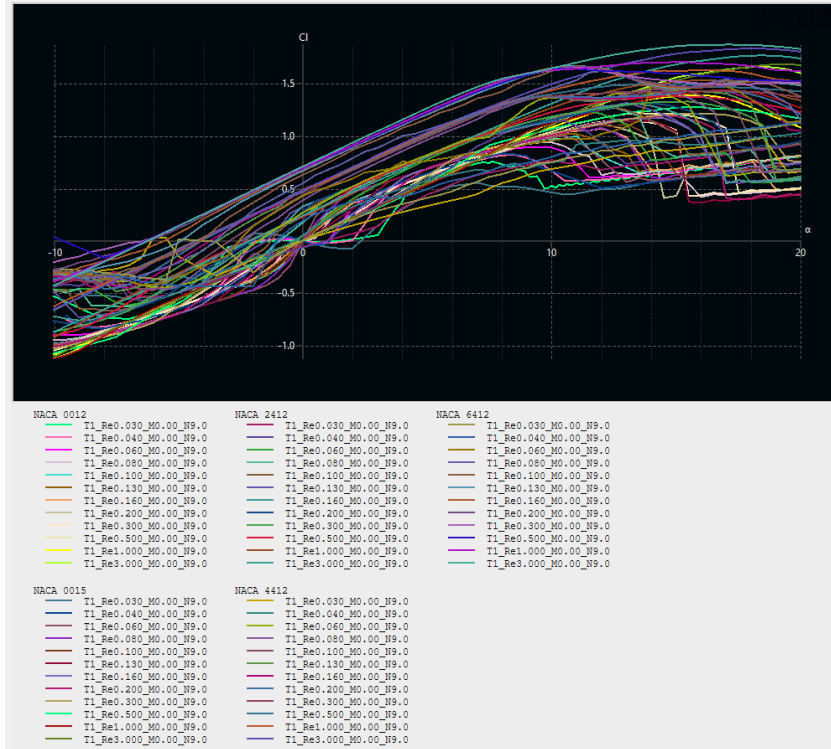


Figure 2: Coefficient of Lift Vs Angle of Attack Graph

Airfoil	Origin	Lift Slope (Per Radian)	0 Lift AoA (Deg)
NACA 0012	Theoretical (Thin Airfoil Theory)	6.28	0.00°
	Numerical (XFLR5)	5.92	0°
NACA 0015	Theoretical (Thin Airfoil Theory)	6.28	0.00°
	Numerical (XFLR5)	6.11	0°
NACA 2412	Theoretical (Thin Airfoil Theory)	6.28	-2.00°
	Numerical (XFLR5)	6.07	-2.24°
NACA 4412	Theoretical (Thin Airfoil Theory)	6.28	-4.00°
	Numerical (XFLR5)	5.96	-4.32°
NACA 6412	Theoretical (Thin Airfoil Theory)	6.28	-6.00°
	Numerical (XFLR5)	5.96	-6.34°

Table 2.2: 2D Airfoil performance parameters

2.2) Why Analytical Theory and Numerical Results Differ

Analytical methods such as thin aerofoil theory or lifting line theory rely on the assumptions that treat air as frictionless and present wings as flat, isolated shapes. In contrast to this, software such as XFLR5 provides a more realistic representation of the aerodynamic behaviour, accounting for flow effects and accounting for the three-dimensional properties of the glider. The main source of discrepancy is the influence of the boundary layer, a thin layer of slower-moving air along the surface of the aerofoil, which can impact the efficiency of lift generation. As a result of this, the slopes generated on XFLR5 displays a value lower than 6.28 (the theoretical lift curve slope per radian from thin aerofoil theory). Another factor that may add to the difference is the aerofoil thickness. Traditional aerofoil analysis assumes a 0-thickness aerofoil to simplify calculations. An addition of thickness to the aerofoil alters the pressure distribution and flow acceleration over the surface, resulting in small discrepancies. This can lead to a more negative 0 lift angle of attack than when calculated numerically. Analytical methods also do not account for 3D aerodynamic interactions and drag mechanisms. Effects such as interference from the main wing and stabilisers, downwash and skin-friction drag go completely unaccounted. Numerical simulations account for all these factors and more, resulting in a more realistic prediction of lift and drag performance.

To validate this further, a comparison was done against the NACA Technical Report No 563. The report consists of lab experiments, force-balance sweeps and pressure tap distributions to verify an experimental 0-lift angle of attack which lies between -4.1° and -4.2° , which falls near the values found through the numerical simulations on XFLR5. The lift slope for the experimental report also trails behind the value found through analysis, again matching the values found through the numerical analysis. The close correlation of the experimental and numerical results provides evidence for the simulation matching the behaviour of real-world aerodynamic behaviours.

2.3) Definitive Airfoil Selection

After a thorough design and validation phase, NACA 4412 was chosen as the superior primary wing airfoil choice. On one hand, NACA 6412 yields a substantial maximum lift; its high camber profile invites severe pitching moments and drag penalties, which ultimately compromise the glider. On the other hand, NACA 2412 provides a low camber option, which eliminates the threat of severe pitching moments that would occur. However, this airfoil significantly underperforms when it comes to low-speed lift capabilities. The NACA 4412 provided a compromise between the two and prevails as an ideal mix between the two airfoils. It introduces good lift boundaries and still eliminates the severe pitching issues and provides well-oriented flight dynamics over a range of Reynolds numbers.

The elevator was designed using NACA 4412 rather than a symmetrical aerofoil such as NACA 0012 or NACA 0015. This was done to improve the directional stabilisation of the glider during its flight. The camber generates correctional forces as the glider undergoes a pitch disturbance, resulting in greater flight stability when compared to a symmetrical aerofoil.

3) Wing Analysis

Parameter	Design Value	Aerodynamic Justification
Wingspan	0.6 m	Maximises aspect ratio within experimental geometry boundaries to directly lower induced drag.
Root Chord	0.090 m	Provides the requisite structural volume and depth to carry bending moment distributions.
Tip Chord	0.090 m	Maintains a constant chord arrangement to eliminate platform curvature manufacturing complexities.
Wing Area	0.054 m^2	Sized explicitly to yield the targeted total vertical lift force at the design velocity of 10 m/s.
Aspect Ratio	6.67	Optimises aerodynamic efficiency and limits induced drag from wingtip vortices within the strict experimental geometric boundaries, balancing aerodynamic performance with structural rigidity.
Taper Ratio	1.000	Rectangular baseline profile ensures high manufacturing repeatability and uniform structural support rib dimensions.
Dihedral	5.0°	Induces passive lateral stability. During a sideslip event, the upward-angled geometry forces the lower wing to experience a higher local angle of attack, creating a restoring rolling moment that automatically returns the glider to a level flight state.
Sweep Angle	0.0°	Eliminates complex spanwise structural-torsional coupling modes typical of swept plans at low Reynolds scales.
Vertical Wing Placement	+1.0 cm (High-Wing)	Establishes a high-body configuration to leverage the pendulum effect, lowering the aircraft's centre of gravity relative to the centre of lift to enhance passive lateral stability. It also isolates the upper-surface root airflow from the fuselage boundary layer to minimise multi-surface interference drag.

Total Lift Force (N)	Total Drag Force (N)
3.313	0.036

TABLE 3.1: TOTAL FORCES COMPUTED BY LIFTING LINE THEORY

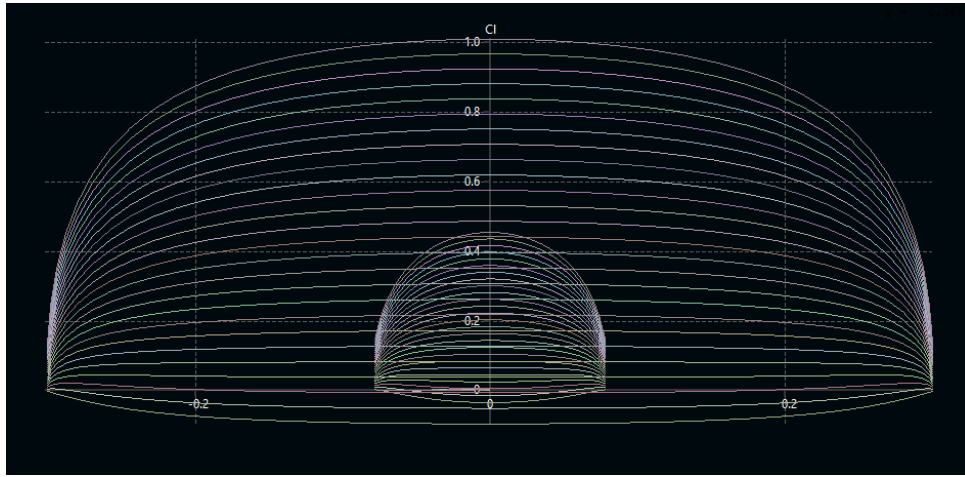


FIGURE 3: THE LIFT DISTRIBUTION CALCULATED USING LIFTING LINE THEORY.

A review of the spanwise aerodynamic data highlights a correlation between the data while also highlighting distinct variations between the lifting line theory and the XFLR5. When integrated along the 60cm wingspan, the lifting line method yields a total lift force of 1.657 N and a total induced drag force of 0.019. However, the XFLR5 yields a lower total lift force 1.150N but a higher total drag force of 0.082N. These variations can be attributed to three main phenomena. First, the manual calculations treat the trailing wake as a finite set of rigid vortex lines. This type of approximation focuses the creating local downwash peaks that spike in sections where XFLR5 would show a smooth continuous line. Second, the multi-surface fin interferences reduce the lift coefficient across the centre of the wing, as seen by **figure 3** where an arch cutout appears. Another reason for this could be that the lifting line theory assumes that the air is frictionless or inviscid. This assumption means that factors such as 3D induced drag, viscous skin friction is not accounted for. These three factors are the main contributors to the discrepancies between the Lifting Line Theory and simulations completed on XFLR5.

4) Glider Design

Subsequent to the completion of the wing analysis, the final design and construction of the glider commenced. As the main wing was designed to be a full three-dimensional aerofoil this required careful consideration as to what materials and overall structure would be used to maintain the wings shape. The initial materials gained for the glider included:

1 x stick of balsa wood, 1 x sheet of foam, ~10 x rubber bands

A solid aerofoil was immediately ruled out with the provided materials as it would have been greatly overweight in comparison to other potential options, as well as requiring glue which would not dissolve the foam, further adding weight and requiring external sourcing. After some consideration, it was decided that the structure of the wing would consist of ribs and spars with a thin material overlayed/stretched over the top to form the lifting surface. This type of structure draws obvious inspiration from the wing structure found on most aircraft (shown in Fig) as it maintains structural strength while still being light.

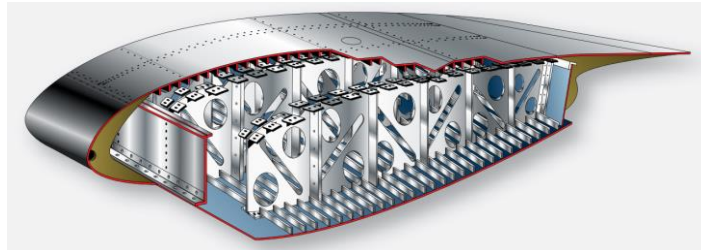


Fig. 4: Wing cross-section showing ribs and spars. Adapted from [2]

Initial construction consisted of printing out 25 2D aerofoils and then cutting them to the lines. This was done using a scalpel, so the cuts were smooth and didn't tear the paper. Afterwards the cut pieces of paper were attached to the foam sheet using a glue stick and then the section containing the attached aerofoils was cut off from the main foam sheet. The foam was then cut around the paper guides thus producing the main shape of the wing ribs. Once this was done two sets of spars (4 in total, 2 smaller, 2 larger) were cut from the foam sheet to the appropriate length, following this 22 of the ribs (11 for each wing) had two slots cut, one in the front and another in the back, with each spar only having one set of slots cut, afterwards the ribs and spars were connected together with a foam rated cyanoacrylate glue. In this stage toothpicks were also inserted near the leading edge of the ribs to better align them. The 3 remaining ribs were joined together to form a single piece which the two wings were attached. Later, this was reinforced due to a minor structural failure. After this the skin was wrapped over the wing, the skin itself was mulberry paper this paper was chosen due to its superior strength and lighter weight compared to traditional paper. The skin was cut into strips and applied over two ribs with PVA glue used to adhere it to the wing structure. With the wing done the rest of the glider was built, simple flat panels were cut out of the foam for the tail section and attached to the body along with the wing where a small foam step was placed before the wing to increase the angle of attack of the wing relative to the body. Finally, a front mass in the form of Blu Tack was placed on the front of the body to move the centre of mass forward (and ensure it was ahead of the centre of lift).



Fig. 5: Completed Glider after construction

5) Flight Evaluation

The final Glider Flight Test showed promising results, performing highly in the areas of flight stability and overall balance. However, the sink rate of the design affected the airborne flight time of the glider, resulting in an average flight time of 1.67s in comparison to the leading team's 5.2s. The recorded results showed an overall performance that was less than expected with regards to our aerodynamic analysis, which prioritised a high L/D ratio and minimal pitching moment effects. These performance criteria directly affected the recorded results in addition to the “chucking” technique and start position of the glider, which may have had room for improvement for greater airborne time.

The post flight analysis unveiled weaknesses in the original horizontal stabilizer and elevator designs, which attributed to the increased pitching moment resulting in a greater sink rate. The effects were mostly visible during the deceleration period of the glider, with the nose pitching forward after reaching stall. Furthermore, the weight distribution may have been further evaluated, repositioning the centre of gravity in an effort to reduce the effects of the pitching moment of the glider.

For future iterations, the findings suggest having the horizontal stabilizer reflex upwards to generate more lift may result in the glider maintaining level flight over a greater period. In addition to this, the centre of gravity may be repositioned using the weight distribution to combat the pitching effects observed during the final test. Finally, the method for construction could be improved as the properties of the material, notably the lack of rigidity in the paper, induced further drag effects during flight.

6) Concluding Remarks

This report documented the design, construction and evaluation of the glider with the primary objectives of achieving the greatest possible airtime whilst also maintaining a balanced flight. Following the evaluation of various airfoils with a high L/D ratio high lift-to-drag ratio and flight stability, geometrical and computational analysis enabled the selection of the NACA 4412 as the chosen airfoil. However, the aircraft underperformed during the final flight test, exhibiting stable flight but only recording an average airtime of 1.67 seconds compared to the best team's 5.2 seconds.

The main cause of the underperformance was attributed to the sink rate and centre of gravity. The relatively low flight time could further be attributed to the tail and elevator design leading the nose of the glider to pitch down, consequentially increasing the sink rate. This occurred when the glider decelerated and therefore inducing stall where the nose pitched forward, thus greatly accelerating the sink rate.

Discrepancies in the expected and predicted performance of the glider could be attributed to the applications utilised in the analytical evaluation, with XFLR5 neglecting 3D flow effects, induced drag and viscous friction effects. In test conditions, the calculations we utilised are unable to fully represent the real effects on the glider. Notably, the 2D hand calculations consistently gave better predicted performance than using software such as XFLR5.

For future iterations of the glider, redesigning the horizontal stabilisers to shift the centre of gravity rearward to the tail of the glider may assist in reducing the pitching moment effects. This would simultaneously improve the overall lift, thus increasing aerodynamic efficiency and flight time.

References

Insert your references here using the IEEE referencing style.

- 1] T. Tuncer, "NACA 4412 Airfoil Data," Department of Aerospace Engineering, Middle East Technical University, Ankara, Turkey. [Online]. Available: <https://ae.metu.edu.tr/tuncer/ae443/docs/NACA4412.pdf>. [Accessed: 8-Jun-2026].
- [2] Federal Aviation Administration and Asa, Aviation Maintenance Technician Handbook-airframe, Ebundle, vol. 1. Aviation Supplies & Academics, 2013, pp. 44 (1-18). Accessed: Jun. 08, 2026. [Online]. Available: <https://www.abbottaerospace.com/downloads/faa-h-8083-31-2-aircraft-maintenance-handbook-vol-1/>

Appendix A – Lifting Line Calculation

Parameter	Value	Unit
2D lift curve slope (a_0)	5.96	1/rad
Reference angle of attack (α_{ref})	5	deg
Twist factor (C)	0	deg
Velocity (V)	10	m/s
Mean chord ($\bar{c}$)	0.090	m
wingspan (b)	0.600	m
Taper ratio (λ)	1.000	
Air Density (ρ)	1.225	kg/m ³

TABLE A.1: INPUT PARAMETERS USED FOR LIFTING LINE ANALYSIS

Y_{CPi}	Value	Y_{Ri}	Value	b_{Ri}	Value	$Chord_{CPi}$	Value	$\alpha_{geomCPi}$	Value	Q_{ij}	Value
Y_{CP1}	0.0118	Y_{R1}	0.0000	b_{R1}	0.0235	$Chord_{CP1}$	0.090	$\alpha_{geomCP1}$	5	Q_{ij1}	7.14
Y_{CP2}	0.0351	Y_{R2}	0.0235	b_{R2}	0.0232	$Chord_{CP2}$	0.090	$\alpha_{geomCP2}$	5	Q_{ij2}	7.24
Y_{CP3}	0.0580	Y_{R3}	0.0468	b_{R3}	0.0226	$Chord_{CP3}$	0.090	$\alpha_{geomCP3}$	5	Q_{ij3}	7.42
Y_{CP4}	0.0802	Y_{R4}	0.0693	b_{R4}	0.0217	$Chord_{CP4}$	0.090	$\alpha_{geomCP4}$	5	Q_{ij4}	7.72
Y_{CP5}	0.1015	Y_{R5}	0.0911	b_{R5}	0.0206	$Chord_{CP5}$	0.090	$\alpha_{geomCP5}$	5	Q_{ij5}	8.16
Y_{CP6}	0.1216	Y_{R6}	0.1116	b_{R6}	0.0192	$Chord_{CP6}$	0.090	$\alpha_{geomCP6}$	5	Q_{ij6}	8.84
Y_{CP7}	0.1403	Y_{R7}	0.1308	b_{R7}	0.0176	$Chord_{CP7}$	0.090	$\alpha_{geomCP7}$	5	Q_{ij7}	9.88
Y_{CP8}	0.1575	Y_{R8}	0.1483	b_{R8}	0.0158	$Chord_{CP8}$	0.090	$\alpha_{geomCP8}$	5	Q_{ij8}	11.60
Y_{CP9}	0.1730	Y_{R9}	0.1641	b_{R9}	0.0137	$Chord_{CP9}$	0.090	$\alpha_{geomCP9}$	5	Q_{ij9}	14.88
Y_{CP10}	0.1866	Y_{R10}	0.1778	b_{R10}	0.0115	$Chord_{CP10}$	0.090	$\alpha_{geomCP10}$	5	Q_{ij10}	24.74

TABLE A.2: LIFTING LINE METRICS

Q	i=1	i=2	i=3	i=4	i=5	i=6	i=7	i=8	i=9	i=10
j=1	7.14	-2.30	-0.52	-0.22	-0.12	-0.08	-0.04	-0.04	-0.02	-0.02
j=2	-0.76	7.24	-2.38	-0.56	-0.24	-0.12	-0.08	-0.04	-0.04	-0.02
j=3	-0.20	-0.80	7.42	-2.52	-0.60	-0.26	-0.14	-0.08	-0.06	-0.02
j=4	-0.10	-0.22	-0.86	7.72	-2.72	-0.68	-0.30	-0.16	-0.08	-0.04
j=5	-0.06	-0.10	-0.26	-0.96	8.16	-3.02	-0.80	-0.34	-0.18	-0.08
j=6	-0.04	-0.06	-0.12	-0.30	-1.10	8.84	-3.48	-0.96	-0.42	-0.18
j=7	-0.02	-0.04	-0.08	-0.14	-0.36	-1.30	9.88	-4.22	-1.26	-0.52
j=8	-0.02	-0.02	-0.04	-0.08	-0.16	-0.44	-1.66	11.60	-5.56	-1.90
j=9	-0.02	-0.02	-0.02	-0.04	-0.08	-0.18	-0.56	-2.42	14.88	-8.62
j=10	0.00	-0.02	-0.02	-0.02	-0.04	-0.08	-0.22	-0.84	-4.66	24.74

TABLE A.3: Q_{II} MATRIX

P	i=1	i=2	i=3	i=4	i=5	i=6	i=7	i=8	i=9	i=10
j=1	7.33	-2.30	-0.52	-0.22	-0.12	-0.08	-0.04	-0.04	-0.02	-0.02
j=2	-0.76	7.43	-2.38	-0.56	-0.24	-0.12	-0.08	-0.04	-0.04	-0.02
j=3	-0.20	-0.80	7.61	-2.52	-0.60	-0.26	-0.14	-0.08	-0.06	-0.02
j=4	-0.10	-0.22	-0.86	7.91	-2.72	-0.68	-0.30	-0.16	-0.08	-0.04
j=5	-0.06	-0.10	-0.26	-0.96	8.35	-3.02	-0.80	-0.34	-0.18	-0.08
j=6	-0.04	-0.06	-0.12	-0.30	-1.10	9.03	-3.48	-0.96	-0.42	-0.18
j=7	-0.02	-0.04	-0.08	-0.14	-0.36	-1.30	10.07	-4.22	-1.26	-0.52
j=8	-0.02	-0.02	-0.04	-0.08	-0.16	-0.44	-1.66	11.79	-5.56	-1.90
j=9	-0.02	-0.02	-0.02	-0.04	-0.08	-0.18	-0.56	-2.42	15.07	-8.62
j=10	0.00	-0.02	-0.02	-0.02	-0.04	-0.08	-0.22	-0.84	-4.66	24.93

TABLE A.4: P MATRIX

β	Value	Γ_i	Value	Γ_{cpj}	Value	L'_{CPj}	Value	$\alpha_{induced j}$	Value	D'_{CPj}	Value
$\beta 1$	0.0873	Γ_{i1}	0.3429	Γ_{cpj1}	0.3411	L'_{CPj1}	4.178	$\alpha_{induced j1}$	0.57°	D'_{CPj1}	0.0413
$\beta 2$	0.0873	Γ_{i2}	0.3375	Γ_{cpj2}	0.3323	L'_{CPj2}	4.071	$\alpha_{induced j2}$	0.59°	D'_{CPj2}	0.0421
$\beta 3$	0.0873	Γ_{i3}	0.3265	Γ_{cpj3}	0.3151	L'_{CPj3}	3.860	$\alpha_{induced j3}$	0.63°	D'_{CPj3}	0.0426
$\beta 4$	0.0873	Γ_{i4}	0.3094	Γ_{cpj4}	0.2902	L'_{CPj4}	3.860	$\alpha_{induced j4}$	0.68°	D'_{CPj4}	0.0424
$\beta 5$	0.0873	Γ_{i5}	0.2858	Γ_{cpj5}	0.2586	L'_{CPj5}	3.168	$\alpha_{induced j5}$	0.74°	D'_{CPj5}	0.0411
$\beta 6$	0.0873	Γ_{i6}	0.2553	Γ_{cpj6}	0.2209	L'_{CPj6}	2.706	$\alpha_{induced j6}$	0.82°	D'_{CPj6}	0.0385
$\beta 7$	0.0873	Γ_{i7}	0.2177	Γ_{cpj7}	0.1783	L'_{CPj7}	2.184	$\alpha_{induced j7}$	0.90°	D'_{CPj7}	0.0343
$\beta 8$	0.0873	Γ_{i8}	0.1725	Γ_{cpj8}	0.1317	L'_{CPj8}	1.613	$\alpha_{induced j8}$	1.00°	D'_{CPj8}	0.0282
$\beta 9$	0.0873	Γ_{i9}	0.1197	Γ_{cpj9}	0.0825	L'_{CPj9}	1.011	$\alpha_{induced j9}$	1.11°	D'_{CPj9}	0.0197
$\beta 10$	0.0873	Γ_{i10}	0.0594	Γ_{cpj10}	0.0322	$L'_{CPj'10}$	0.394	$\alpha_{induced j10}$	1.25°	D'_{CPj10}	0.0086

TABLE A.5: LIFTING LINE MATRICES**Appendix B – Equations**

$$M = e^{\frac{t}{0.482 \times T}} + 2 \quad (1)$$

$$L = C_L \frac{1}{2} \rho V^2 A \quad (2)$$

$$D = C_D \frac{1}{2} \rho V^2 A \quad (3)$$